

Elliot MacKrell, Ph.D.

emackrel@caltech.edu • Pasadena, CA 91106 • (954) 651-2968

[GitHub Portfolio](#) • [Personal Website](#) • [LinkedIn](#)

Education

California Institute of Technology

Pasadena, CA

Ph.D. Chemical Engineering

2016-2024

Thesis (Defended 6/20/24): *Global Analysis of Protein Synthesis and Degradation in Escherichia coli*

University of Florida

Gainesville, FL

B.S. Chemical Engineering

2016

Minor: Biomolecular Engineering

Research Experience

California Institute of Technology, Division of Chemistry and Chemical Engineering

2016-Present

Postdoctoral Scholar (2024-Present); Ph.D. Candidate - NSF Graduate Fellow (2016-2024); Advisor: Prof. David A. Tirrell

- Discovered global regulatory roles of proteolysis by combining chemoproteomic analysis, linear modeling, text mining, and machine learning to identify 189 novel protease substrates in *Escherichia coli*.
- Developed a predictive machine learning model for *in vivo* protein stability in *E. coli*, demonstrating high accuracy (87.5%) with experimental validation of 8 substrates.
- Developed and deployed an open-source interactive application for filtering, normalization, differential expression analysis, and pathway enrichment analysis of mass spectrometry-based proteomics data. ([repository](#)) ([publication](#))
- Identified DNA repair factors recruited to colibactin-induced DNA interstrand cross-links in *Xenopus* egg extracts via differential expression analysis and time-series clustering of chromatin mass spectrometry data. ([preprint](#))

EMD Serono, Department of Translational Medicine and Global Clinical Development

2022

Bioinformatics Co-op; Supervisor: Dr. Irina Kalatskaya

- Developed Shiny application for visualizing cell type-specific transcript abundance, identifying differentially expressed genes, and conducting gene set enrichment analysis in clinical samples across autoimmune disease states.
- Established pipeline and Shiny application for the real-time analysis of text mining data from PubMed to identify genes, pathways, and diseases of therapeutic interest through information-theoretic and statistical approaches.

Harvard Medical School, Wellman Center for Photomedicine, Massachusetts General Hospital

2015

NSF REU Fellow; Advisor: Prof. Robert Redmond

- Investigated the effects of photocrosslinking on ex vivo tissue biomechanics via cyclic uniaxial tensile testing.

Johns Hopkins University School of Medicine, Department of Radiation Oncology

2014

NSF REU Fellow; Advisor: Prof. Robert Ivkov

- Evaluated the anti-tumor immune response generated by magnetic nanoparticle hyperthermia and ionizing radiation treatments in a mouse model for cancer.

University of Florida, Department of Chemistry

2012-2015

Undergraduate Research Assistant; Advisor: Prof. Brent S. Sumerlin

- Developed an amphiphilic and pH-responsive copolymer for phloem-specific drug delivery in plants by functionalizing polysuccinimide (PSI) with various hydrophilic molecules.

Publications

Altshuller, M., He, X., **MacKrell, E. J.**, Wernke, K. M., Wong, J.W. H., Sellés Baiget, S., Wang, T. Y., Chou, T. F., Duxin, J. P., Balskus, E. P., Herzon, S. B., Semlow, D. R. "The Fanconi anemia pathway repairs colibactin-induced DNA interstrand cross-links." **bioRxiv**, 2024. <https://doi.org/10.1101/2024.01.30.576698>

Jones, J.J., **MacKrell, E.J.**, Wang, T.Y., Lomenick, B., Roukes, L.M., and Chou, T.F. “tidyproteomics: An open-source R package and data object for quantitative proteomics post analysis and visualization.” **BMC Bioinformatics**, **2023**, *24*, 239. <https://doi.org/10.1186/s12859-023-05360-7>

Hill, M.R., **MacKrell, E.J.**, Forsthoefel, C.P., Jensen, S., Chen, M., Moore, G., He, Z.L. and Sumerlin, B.S. “Biodegradable and pH-Responsive Nanoparticles Designed for Site-Specific Delivery in Agriculture.” **Biomacromolecules**, **2015**, *16*, 1276-1282. <https://doi.org/10.1021/acs.biomac.5b00069>

Manuscripts in preparation

MacKrell, E. J., Lomenick, B., Chou, T. F., and Tirrell, D. A. “Global protease substrate identification reveals instability of diverse regulators in *Escherichia coli*.” Manuscript in preparation.

MacKrell, E. J., Lomenick, B., Chou, T. F., and Tirrell, D. A. “MazF activates the cold shock response in *Escherichia coli*.” Manuscript in preparation.

Technical Skills

Bioinformatics and statistics: Mass spectrometry-based proteomics analysis, RNA-seq analysis, Machine learning, Text mining, Cloud computing, Shiny application development, Linear modeling, Regulatory network analysis, Gene set enrichment, Bayesian inference

Programming languages: Python, R, SQL, Bash, JavaScript, CSS, HTML

Machine learning: Scikit-learn, PyTorch, XGBoost

Data analysis and visualization: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Shiny, Bioconductor, tidyverse

Cloud and development tools: Docker, AWS, Git, Visual Studio Code, RStudio

Software suites: GeneData Profiler, Proteomics analysis suites (MaxQuant, Proteome Discoverer), Clarivate CBDD

Laboratory skills: Molecular cloning, Bacterial culture, Genome editing (Recombination, CRISPR-Cas9), Immunoblotting, Confocal microscopy, Proteomics sample preparation

Awards and Honors

CEMI Pilot Grant , California Institute of Technology	2018
CEMI Fellow , California Institute of Technology	2016-2024
NSF Graduate Research Fellowship , California Institute of Technology	2016-2021
Yao Su Student Research Prize , HST Summer Institute	2015
NSF REU Fellowship , Harvard-MIT Division of Health Sciences and Technology (HST)	2015
Collins Engineering Scholarship , University of Florida	2015
NSF REU Fellowship , Johns Hopkins University Institute for NanoBioTechnology (INBT)	2014
University Scholars Program , University of Florida	2013